# CUSTOMER CHURN ANALYTICS

Business Problem Statements & Project Deliverables

## Project Overview

This project uses Python, MySQL, and Power BI to prepare and analyze customer data, explore churn-related patterns, and present findings through an interactive dashboard. The cleaned dataset contains **492 records and 23 columns**.

## 1. Business Problem Statements

- **Understand customer churn:** What proportion of customers have churned, and what does the overall churn rate indicate?
- **Examine customer value:** Which customers have the highest customer value, and how can customer value help prioritize retention analysis?
- **Compare contract types:** How are customers distributed across contract types, and what patterns should the business investigate further?
- **Explore customer retention:** What patterns in the available customer attributes may help the business understand customer retention and potential churn?
- **Support data-driven decisions:** How can the cleaned data and dashboard help stakeholders monitor churn-related indicators and identify areas for further investigation?

## 2. Project Deliverables

- **Cleaned customer dataset:** A processed CSV file containing 492 records and 23 columns, ready for analysis.
- **Python notebook:** A Jupyter Notebook documenting data loading, inspection, cleaning, and preprocessing.
- **SQL analysis script:** A MySQL script containing queries for churn rate, customer value, contract-type distribution, and related analysis.
- **Power BI dashboard:** An interactive dashboard presenting customer churn and customer-related indicators.
- **Project documentation:** A README file describing the project overview, tools, workflow, files, and execution steps.

## 3. Project Files

| File | Purpose |
|---|---|
| Churn Dataset Cleaning.ipynb | Python data cleaning and preprocessing notebook |
| Churn_Unclean_Project.xlsx | Original source dataset |
| Customer_Churn_Cleaned.csv | Cleaned dataset for analysis |
| Customer_Churn_Analysis.sql | MySQL analysis queries |
| Customer_Churn_Analysis_Dashboard.pbix | Power BI dashboard |

## 4. Expected Business Value

The analysis is intended to help stakeholders understand churn-related measures, review customer value and contract distribution, and use the dashboard as a starting point for customer-retention discussions. The project does not claim causal drivers or predictive churn performance unless those are separately tested and validated.

Prepared by: Akash Pal | Tools: Python, MySQL, Power BI